

Evaluation of Machine Learning methods for Detection of Ventricular Arrhythmia using HRV Metrics

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Abstract—This paper evaluates three machine learning classifiers for automated detection of ventricular tachycardia and ventricular fibrillation (VT/VF) using heart rate variability (HRV) features derived from RR intervals. A Support Vector Machine, an XGBoost ensemble, and a Multi-Layer Perceptron are each trained across six hyperparameter configurations on the MVTDB PhysioNet dataset. The best-performing configuration of each classifier is selected by F1-score and compared on accuracy, precision, recall, and F1. The effect of Butterworth low-pass filtering in the preprocessing pipeline is assessed by re-evaluating each best configuration on unfiltered data. Results show that class imbalance is the dominant factor in classification performance, and that loss-function asymmetry consistently produces larger gains than architectural changes. The MLP achieves the highest recall (0.9163) and F1 (0.6480) with class weighting but at low precision (0.5012). The SVM achieves the highest accuracy (0.6810) and precision (0.7261) but the lowest recall (0.5022). XGBoost offers the most balanced error profile for clinical deployment. Butterworth filtering improves all classifiers on all metrics by 2-8 percentage points without changing their relative ranking.

Index Terms—ECG Signals, Signal Processing, Machine Learning, Support Vector Machine, XGBoost, Gradient Boosting, Multi-Layer Perceptron, Heart Rate Variability

I. INTRODUCTION

Ventricular tachycardia (VT) and ventricular fibrillation (VF) are life-threatening cardiac arrhythmias that, if left undetected can lead to sudden collapse, followed by death due to lack of blood flow to the brain and lungs. Early and reliable detection is therefore of significant clinical importance, as quick intervention is critical for patient survival.

Heart Rate Variability (HRV), derived from the RR-interval of an ECG signal, provides a set of time-domain and frequency-domain features that reflect autonomic nervous system activity in the heart. These metrics differ substantially between normal sinus rhythm and arrhythmia, making them a suitable set of features for training machine learning algorithms.

This paper evaluates three machine learning classifiers in a variety of configurations for binary VT/VF detection using HRV features:

- Support Vector Machines (SVM)
- XGBoost models, a Gradient Boosting ensemble method
- Multi-Layer Perceptron (MLP) neural networks

The classifiers are trained on the MVTDB RRdata1 dataset from PhysioNet, which contains paired recordings of arrhythmias and baseline rhythms from the same patients. For each classifier, multiple hyperparameter configurations were trained and compared. The best-performing configurations were then compared across classifiers on their accuracy, precision, recall, and F1-score to assess which is most appropriate for the task. Finally, the effect of low-pass Butterworth noise filtering in the preprocessing pipeline is examined by re-evaluating each best configurations on unfiltered RR windows and comparing the results before and after filtering.

II. HYPERPARAMETER PERFORMANCE COMPARISON

Each classifier was evaluated across six hyperparameter configurations, with the best-performing configuration selected by validation F1-score. F1-score is used here as a balanced overall metric, as it accounts for both precision and recall rather than rewarding a classifier that simply predicts the majority class. In a clinical setting, recall can be a more appropriate choice of primary metric, since a false negative represents a missed VT/VF event which could be fatal. A false positive, while undesirable, carries far less risk. Section 3 analyses why, in practice, prioritising recall over F1 score can have unintended consequences. The following subsections present the configurations tested for each classifier and the results obtained.

A. Support Vector Machine

The two primary hyperparameters of an SVM are the kernel and the regularisation parameter C. The kernel determines the shape of the decision boundary: a linear kernel separates classes with a straight hyperplane, while radial (RBF) and polynomial kernels implicitly map the features into higher-dimensional spaces to capture more complex, non-linear boundaries. The C parameter controls how strictly the model

enforces correct classification of training examples. A low C permits some misclassifications in exchange for a wider, more generalised margin, while a high C shrinks the margin to minimise training errors, at the risk of overfitting.

1) *Configurations*: The first four configurations explore the three available kernel types and the effect of class balancing. Results from these showed that class balancing improved recall, and that the linear kernel was competitive with RBF despite its simplicity. Configs 5 and 6 build on these observations. Config 5 combines the RBF kernel with balanced weighting and raises C to 10, pushing the model to correctly classify more hard positive examples at the cost of a narrower margin. Config 6 applies the same high- C balanced strategy to the linear kernel, testing whether a simpler boundary can be made recall-aggressive without the additional capacity of a non-linear kernel.

TABLE I
SUPPORT VECTOR MACHINE CONFIGURATIONS

Config	Explanation
Simple Linear Kernel	A linear kernel produces a linear decision boundary. A minimally tuned baseline for comparison.
Radial Basis Function Kernel	A radial kernel maps features into higher dimensions to capture a curved decision boundary.
RBF + Balanced Classes	Same as the baseline configuration but loss is reweighted so mis-labelled VT/VF hurts as much as mislabelled baseline rhythm.
Polynomial Kernel	Uses feature interaction terms (i.e. thresholds of multiple features simultaneously) to model more complex decision boundaries.
RBF + Balanced + High C	A high C value aggressively focuses on correct classification of hard positive examples by shortening the SVM margin.
Linear + Balanced + High C	A linear boundary with a high C value heavily penalises misclassification of either class, making the model fast and recall-aggressive.

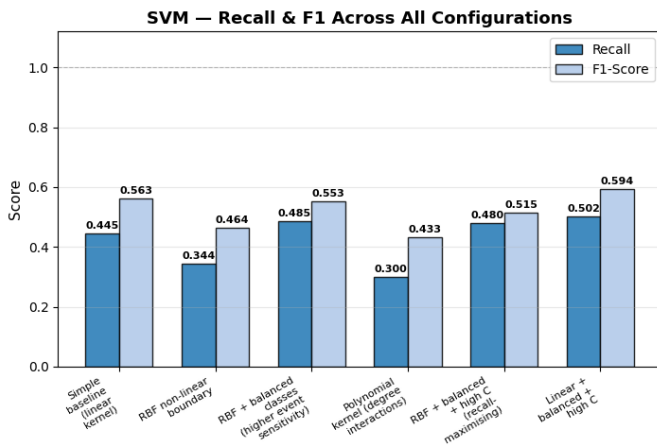


Fig. 1. Support Vector Machine - F1 and Recall Scores

TABLE II
SUPPORT VECTOR MACHINE - RESULTS FOR ALL CONFIGURATIONS

Config	Accuracy	Precision	Recall	F1 Score
Simple Linear Kernel	0.6789	0.7652	0.4449	0.5627
Radial Basis Function Kernel	0.6319	0.7156	0.3436	0.4643
RBF + Balanced Classes	0.6360	0.6433	0.4846	0.5528
Polynomial Kernel	0.6360	0.7816	0.2996	0.4331
RBF + Balanced + High C	0.5808	0.5561	0.4802	0.5154
Linear + Balanced + High C	0.6810	0.7261	0.5022	0.5938

2) *Results*: The linear kernel outperformed both RBF and polynomial kernels across all exploratory configurations, which is notable given that RBF is generally considered the more expressive option. This suggests the HRV feature space is already structured in a way that benefits linear separation, or that the additional complexity of a non-linear boundary introduces noise rather than signal on this dataset size.

Switching from the unweighted linear baseline to RBF actually reduced F1 from 0.5627 to 0.4643, and recall dropped from 0.4449 to 0.3436. Adding balanced class weights to the RBF kernel (Config 3) partially recovered recall to 0.4846, confirming that class imbalance was a significant bottleneck for the non-linear kernels. The polynomial kernel (Config 4) achieved the highest precision of all six configurations (0.7816) but the lowest recall (0.2996), indicating it was strongly biased toward the majority class and rarely predicted VT/VF.

Applying high C with balanced weighting to the RBF kernel (Config 5) did not improve on Config 3, with F1 falling to 0.5154. The same combination applied to the linear kernel (Config 6) produced the best result overall: F1 of 0.5938 and recall of 0.5022. The linear kernel appears to respond well to aggressive margin enforcement in a way that RBF does not, likely because the tighter margin on a linear boundary still generalises, whereas a tighter margin on an already overfitted RBF boundary compounds the problem.

B. XGBoost

XGBoost builds an ensemble of decision trees sequentially, where each tree is trained to correct the residual errors of the previous ones. The number of estimators controls how many trees are in the ensemble: more trees allow the model to correct finer residual errors, but increase the risk of overfitting and training time. The maximum depth limits how many levels each individual tree can have. Shallow trees are faster and less likely to overfit, while deeper trees can capture more complex interactions between features but are more prone to memorising the training data.

1) *Configurations*: The first four configurations vary ensemble size, tree depth, learning rate, and subsampling to explore the parameter space. Results showed that class imbalance was suppressing recall across all configurations, with the model consistently favouring the majority class. Config 5 takes the regularised setup from Config 4 and applies a positive class weight of 3, penalising false

negatives three times as heavily. This directly targets the recall deficiency by shifting the cost asymmetry. Config 6 explores higher model capacity: 500 trees at depth 5 with a slow learning rate of 0.02 and mild subsampling, testing whether a substantially larger ensemble can recover accuracy without sacrificing the regularisation benefits of a controlled learning rate.

TABLE III
XGBOOST CONFIGURATIONS

Config	Explanation
Simple baseline (100 Trees, Depth 3, LR 0.1)	A sane reference point with shallow trees, fewer trees, fast learning rate
Deeper Trees (150 Trees, Depth 5, LR 0.1)	Increased tree depth and tree count to capture more complex feature interactions
More Trees + Lower LR (300 Trees, Depth 4, LR 0.03)	Lower learning rate means many small correction steps instead of fewer large ones, which tends to generalise better
Subsampled + Regularized (200 Trees, Depth 4, LR 0.05, Sub/col-sample 0.8)	Random row and feature sampling per tree helps to reduce overfitting from lower learning rate
Subsampled + Regularized + Positive Class Bias (200 Trees, Depth 4, LR 0.05, Sub/col-sample 0.8, scale_pos_weight=3)	Identical to the above configuration but false negatives are penalised 3x as harshly, which trades some precision for enhanced recall
More Trees + Deeper + Lower LR (500 Trees, Depth 5, LR 0.02, Sub/col-sample 0.85)	A larger model with 500 trees, further depth, and a slower learning rate with some mild subsampling to enhance accuracy

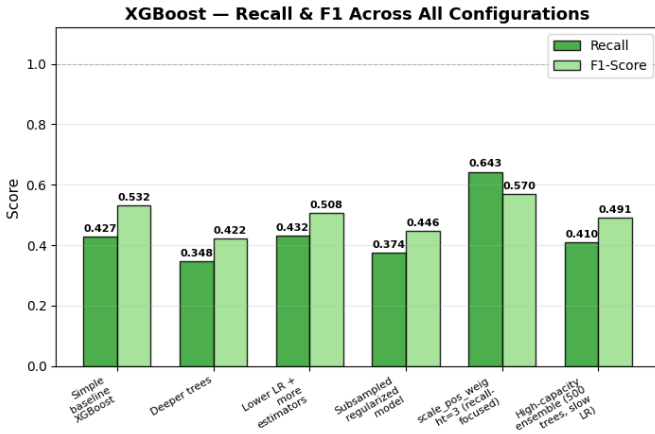


Fig. 2. XGBoost - F1 and Recall Scores

2) *Results*: The shallow baseline (Config 1) outperformed every subsequent architectural and regularisation change in the first four configurations, which points to overfitting as the primary issue. Config 2, despite adding more trees and greater depth, degraded across all metrics relative to the baseline, with F1 falling from 0.5315 to 0.4225. This counterintuitive result suggests that at this dataset size,

TABLE IV
XGBOOST - RESULTS FOR ALL CONFIGURATIONS

Config	Accuracy	Precision	Recall	F1 Score
Baseline (100 Trees, Depth 3)	0.6503	0.7029	0.4273	0.5315
Deeper Trees (150 Trees, Depth 5)	0.5583	0.5374	0.3480	0.4225
More Trees + Lower LR	0.6115	0.6164	0.4317	0.5078
Subsampled + Regularized	0.5685	0.5519	0.3744	0.4462
Regularized + Positive Class Bias	0.5501	0.5123	0.6432	0.5703
More Trees + Deeper + Lower LR	0.6053	0.6118	0.4097	0.4908

deeper trees memorise training patterns rather than learning generalisable ones.

Config 3 partially recovered performance by using a lower learning rate with more trees, but still did not match the baseline. Config 4 added subsampling as a further regulariser, yet continued to underperform. Across all four of these configurations, recall remained in the 0.35-0.43 range, indicating a consistent bias toward the majority class that architectural choices alone could not address.

Config 5 broke this pattern by introducing a positive class weight of 3. Recall jumped to 0.6432, the highest of any XGBoost configuration, and F1 rose to 0.5703, surpassing the baseline. This was a larger improvement than any architectural change across the previous four experiments. Config 6, which pushed capacity further with 500 trees and greater depth, did not build on Config 5's recall gains, confirming that the bottleneck was cost function asymmetry rather than model expressiveness.

C. Multi-layer Perceptron

The two primary structural hyperparameters of an MLP are the number of hidden layers and the size of each layer. Adding more hidden layers allows the network to learn increasingly abstract representations of the input, which can improve performance on complex tasks but also increases the risk of overfitting on smaller datasets. Increasing layer size gives each stage more neurons to work with, which raises the model's capacity to separate classes, but similarly increases the risk of overfitting and slows training. Both parameters interact: a deeper network with small layers may underfit, while a shallow network with very large layers may fail to capture the sequential feature abstractions that depth provides.

1) *Configurations*: The first four configurations hold architecture roughly fixed and vary the optimiser and learning rate schedule. None achieved strong recall, which suggested the class imbalance was not being adequately addressed by these choices alone. Config 5 takes the AdamW plus cosine annealing setup from Config 2 and applies a positive class weight of 3 via the loss function, making missed VT/VF events three times more costly than missed baseline rhythms. Config 6 tests a three-layer network [128, 64, 32] with AdamW and cosine annealing at a lower learning rate of 0.0005, assessing whether greater depth and capacity allow the model to learn more useful feature representations from the HRV data.

TABLE V
MULTI-LAYER PERCEPTRON CONFIGURATIONS

Config	Explanation
Baseline: ADAM, No Scheduler	A single small hidden layer with a fixed learning rate. A very basic model
ADAMW + CosineAnnealing	Two hidden layers with weight decay. Learning rate anneals smoothly to near-zero over the course of training
SGD + StepLR Decay	Stochastic Gradient Descent parameter optimisation with the Learning Rate halved every third of training. A very classical approach to MLP training.
RMSprop + ReduceOnPlateau	A per-parameter Learning Rate combined with automatic LR drop when validation loss stops improving
ADAMW + CosineAnnealing + Positive Class Weighting	Same as configuration 2 but the positive class is weighted 3x higher, meaning false negatives are heavily penalised, promoting high recall
Extra Layer [128, 64, 32] + ADAMW + CosineAnnealing	A network with 3 hidden layers, higher dropout, and smooth LR decay, which provides the highest capacity to model non-linearities in the dataset

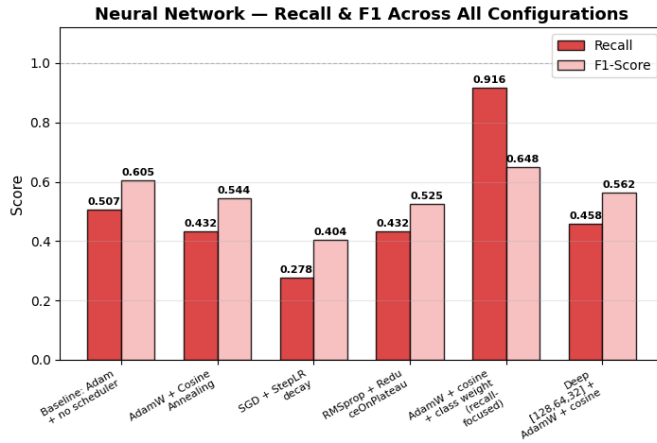


Fig. 3. Multi-layer Perceptron - F1 and Recall Scores

2) *Results*: The simple Adam baseline (Config 1) achieved the strongest performance among the first four configurations, with an F1 of 0.6053 and recall of 0.5066. More sophisticated optimisers did not improve on this: AdamW with cosine annealing reduced recall to 0.4317, and SGD with step decay produced the weakest recall of all six configurations at 0.2775. SGD’s poor recall despite high precision (0.7412) suggests it converges to a solution that largely predicts the majority class, which a fixed step schedule cannot correct for. The adaptive optimisers (Configs 2 and 4) reached identical recall (0.4317) but AdamW edged ahead on F1 due to higher accuracy and precision.

Config 5 produced the most striking result of any configuration across all three classifiers in this study. Applying a positive class weight of 3 to the AdamW plus cosine

TABLE VI
MULTI-LAYER PERCEPTRON - RESULTS FOR ALL CONFIGURATIONS

Config	Accuracy	Precision	Recall	F1 Score
Baseline (ADAM + No scheduler)	0.6933	0.7516	0.5066	0.6053
ADAMW + CosineAnnealing	0.6646	0.7368	0.4317	0.5444
SGD + StepLR Decay	0.6196	0.7412	0.2775	0.4038
RMSprop + ReduceOnPlateau	0.6380	0.6712	0.4317	0.5255
ADAMW + Cosine + Class Weight	0.5378	0.5012	0.9163	0.6480
Deeper Network + ADAMW + Cosine	0.6687	0.7273	0.4581	0.5622

annealing setup raised recall from 0.4317 to 0.9163, which means the model correctly identified over 91% of VT/VF events in the test set. The trade-off was a drop in precision from 0.7368 to 0.5012 and in accuracy from 0.6646 to 0.5378, reflecting a precision-recall trade-off when the loss function is heavily tilted toward one class. The resulting F1 of 0.6480 is the highest of any MLP configuration and the highest of any single classifier in this study. Even with the averaging applied by the F1 Score, the significant jump in recall causes these configurations to pull ahead, despite their dangerously low accuracy and precision, making them effectively useless in a clinical setting due to the high volume of false positives.

Config 6, the deeper three-layer network, improved on most of the unweighted configurations (F1 0.5622) but could not match Config 1’s baseline. The additional capacity did not help in the absence of a corrected cost function. The consistent pattern across both XGBoost and MLP configurations is that loss reweighting produced larger recall gains than any architectural modification.

III. CLASSIFIER PERFORMANCE COMPARISON

Table VII compares the best-performing configuration of each classifier. A notable feature of all three best configurations is that they each required class weighting or cost-asymmetry modifications to reach their peak performance. None of the architecturally varied but symmetrically trained configurations produced the best result for any classifier. This is consistent with the finding across Section 2 that class imbalance was the dominant obstacle in this classification task, and it suggests that any classifier applied to this type of imbalanced medical dataset should treat cost-function design as a first-order concern rather than an afterthought.

The three classifiers differ substantially in how they balance precision and recall. The SVM achieves the highest accuracy (0.6810) and precision (0.7261) but the lowest recall (0.5022). XGBoost sits in the middle on both dimensions. The MLP achieves a recall of 0.9163, far ahead of both alternatives, at the cost of the lowest accuracy (0.5378) and precision (0.5012) of the three. The following subsections analyse what these differences mean in terms of F1 score, error types, and clinical utility.

A. F1 Score

By F1 score, the MLP leads with 0.6480, followed by the SVM at 0.5938, and XGBoost at 0.5703. F1 is the harmonic

TABLE VII
MODEL PERFORMANCE COMPARISON (BEST CONFIGS)

Config	Accuracy	Precision	Recall	F1 Score
SVM - Linear + Balanced + High C	0.6810	0.7261	0.5022	0.5938
XGBoost - Subsamp. + Reg. + Class Weighting	0.5501	0.5123	0.6432	0.5703
MLP - ADAMW + Cosine + Class Weighting	0.5378	0.5012	0.9163	0.6480

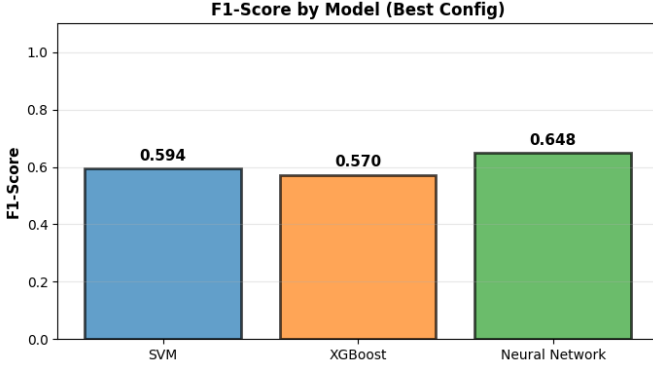


Fig. 4. Comparison of F1 Scores for each ML Model

mean of precision and recall, and it is a useful summary metric precisely because it penalises extreme imbalances between the two. A classifier that predicts positive for every sample would score a recall of 1.0 but a precision equal to the class prevalence. The F1 of such a classifier would be low, which is the intended behaviour of the metric. However, F1 can still be misleading when recall is very high and precision is moderate. In the MLP's case, a recall of 0.9163 is so dominant that it lifts the harmonic mean despite a precision of only 0.5012. The F1 therefore makes the MLP appear approximately 9 percentage points ahead of the SVM, but that gap is driven almost entirely by recall. A model that correctly classifies every VT/VF event but generates an equal number of false alarms may not be clinically useful, yet it scores well on F1. This is why F1 alone is insufficient to evaluate a clinical classifier, and the following subsection examines recall in isolation to understand the tradeoffs more clearly.

B. Recall Score

On recall alone, the MLP (0.9163) is far ahead of XGBoost (0.6432) and SVM (0.5022). In a clinical setting, recall is more directly tied to patient safety than any other metric: a false negative means a VT/VF episode was not detected, which could be fatal. This might suggest that maximising recall should be the primary design goal.

The problem with that reasoning is that recall can always be pushed to 1.0 by predicting positive for every sample. The meaningful question is not which classifier has the highest recall, but which classifier achieves the best recall without generating an unacceptable number of false alarms. The MLP's precision of 0.5012 means that roughly one in every two positive predictions is incorrect. In a continuous monitoring scenario, this would generate a sustained flood

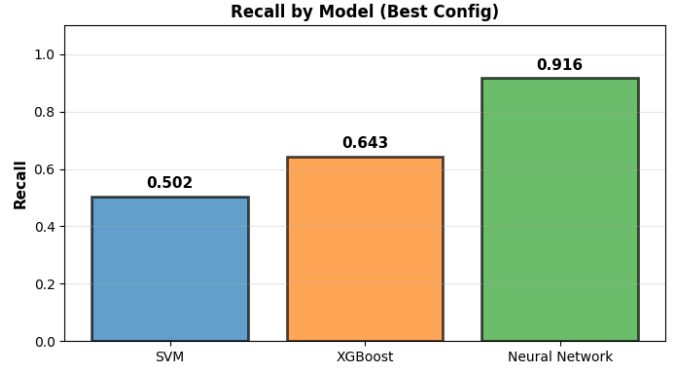


Fig. 5. Comparison of Recall Scores for each ML Model

of false positive alerts. Alarm fatigue is a documented clinical safety risk: when clinical staff are repeatedly alerted to events that are not real, they become desensitised, and genuinely dangerous events are more likely to be dismissed or delayed. A classifier with very high recall but very low precision can therefore reduce patient safety rather than improve it.

XGBoost offers a more tenable balance: recall of 0.6432 with precision of 0.5123. It catches substantially more VT/VF events than the SVM while generating fewer false positives than the MLP. The SVM's recall of 0.5022 means it misses approximately half of all VT/VF events in the test set, which is too low to be reliable as a standalone detection system regardless of its superior precision.

C. Confusion Matrices

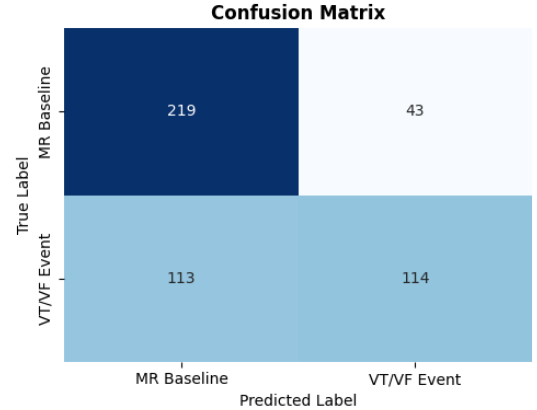


Fig. 6. Confusion Matrix of best Support Vector Machine

The confusion matrices in Figures 6, 7, and 8 reveal the error profiles of each classifier. The SVM has the fewest false positives of the three, consistent with its high precision of 0.7261. However, it produces the most false negatives: it correctly identifies only around half of all VT/VF windows. The confusion matrix reflects a classifier that has learned a conservative boundary, correctly labelling the majority of baseline rhythm windows but regularly missing arrhythmia

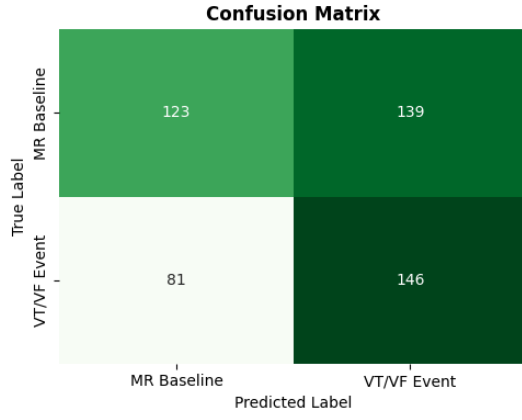


Fig. 7. Confusion Matrix of best XGBoost Model

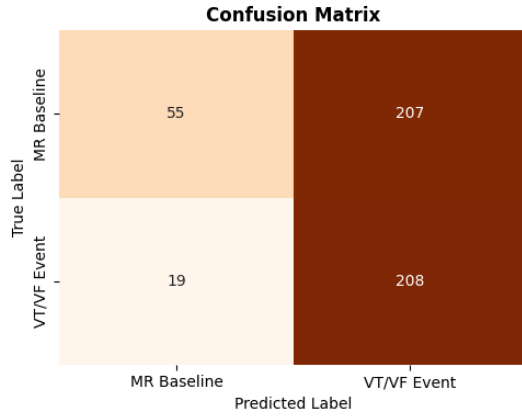


Fig. 8. Confusion Matrix of best Multi-Layer Perceptron Model

events. This is the expected behaviour of a linear SVM: the decision hyperplane is a global, rigid boundary in feature space, and balanced class weighting with high C forces it to be more aggressive but the linear constraint limits how well it can adapt to the specific distribution of the minority class. XGBoost shows fewer false negatives than the SVM but more false positives. This is consistent with the effect of `scale_pos_weight`, which biases each tree's splits toward the VT/VF class. Because XGBoost builds its ensemble by sequential error correction, the class-weighted loss function causes later trees to disproportionately focus on the harder-to-classify positive examples. The result is a classifier that captures more true VT/VF events but at the cost of pulling some borderline baseline windows into the positive class. The MLP confusion matrix shows a dramatically different error profile. False negatives are very few, reflecting the near-complete recall of 0.9163. The dominant error is false positives. With a loss weight of 3 on the positive class applied at every backpropagation step, the network's weights are optimised much more strongly to avoid missing VT/VF events than to avoid incorrect positive predictions. The network is essentially learning to be highly sensitive at the cost of specificity. This is a property of the MLP's

gradient-based training: the loss weighting directly distorts the gradient signal, and the network with many parameters has the capacity to overfit to this asymmetric objective in a way that a linear SVM or a shallow decision tree cannot.

D. Discussion

No classifier in this study achieves both high precision and high recall simultaneously. This precision-recall tradeoff is not a deficiency of a specific model choice; it is a fundamental property of classification under class imbalance, and it means there is no single objectively "best" classifier here. The appropriate choice depends on the deployment context and on what the cost of each error type is considered to be. The SVM, with the highest precision (0.7261) and accuracy (0.6810), is the most conservative classifier. When it predicts VT/VF, that prediction is correct roughly 73% of the time. Its drawback is that it misses approximately half of all true arrhythmia events, which is a clinically significant false negative rate. In a low-alert-volume monitoring system where every alert is immediately reviewed by a clinician, this conservative behaviour may be acceptable. In a setting where missed events carry catastrophic consequences, it is not.

The MLP, while leading on both recall and F1, is essentially unusable as a standalone clinical decision tool in its current form. A precision of 0.5012 means one in every two alerts is a false alarm. In a continuous monitoring scenario this would produce a sustained stream of incorrect positive predictions, contributing to the alarm fatigue problem that already affects ICU environments. The MLP's behaviour reflects what happens when a model is optimised primarily for recall without an explicit constraint on false positive rate.

XGBoost offers the most balanced error profile in the context of clinical deployment. It improves meaningfully on the SVM's recall (0.6432 vs 0.5022) without generating the volume of false positives that the MLP produces. Neither its recall nor its precision is exceptional in isolation, but the combination is the most defensible for a monitoring application where both error types carry real cost.

A broader observation across all three classifiers is that the overall accuracy figures are modest, with the best-performing model on accuracy achieving only 0.6810. This indicates that RR-interval-derived HRV features alone may not fully capture the arrhythmia signature. VT and VF produce distinctive changes in RR intervals, but HRV features derived from those intervals are aggregate statistical summaries that can overlap with the HRV profile of other rhythms. Incorporating waveform morphology features from the raw ECG signal would likely provide additional discriminative power and could improve performance across all classifiers.

IV. CLASSIFIER NOISE RESILIENCE COMPARISON

Each classifier was re-evaluated on the same test split using RR interval windows that were not passed through the

Butterworth low-pass filter, with all other preprocessing and model configuration held constant. Table VIII shows the improvement in each metric when filtering is enabled. All three classifiers improved on every metric with filtering applied, confirming that the filter consistently removes noise that would otherwise degrade classification. The degree and distribution of those improvements, however, differs across classifiers.

TABLE VIII
IMPROVEMENT OF EACH CLASSIFIER WITH THE ADDITION OF FILTERING

Model	Accuracy	Precision	Recall	F1 Score
SVM	+4.50%	+8.11%	+2.20%	+4.32%
XGBoost	+3.68%	+3.29%	+7.93%	+5.21%
MLP	+1.64%	+1.02%	+7.05%	+2.66%

A. F1 Score Comparison

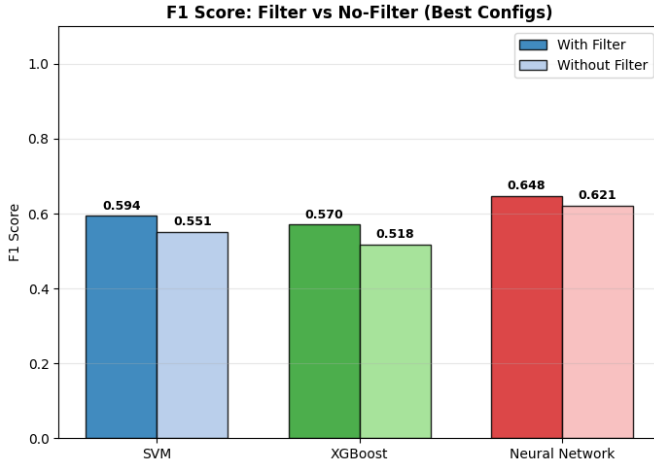


Fig. 9. F1 Score Comparison with and without filtering for each classifier

XGBoost shows the largest F1 improvement with filtering (+5.21%), followed by SVM (+4.32%) and MLP (+2.66%). The smaller gain for the MLP is consistent with its architecture: dropout regularisation during training already provides a degree of noise robustness, and the dominant effect of the class-weighted loss tends to overshadow the contribution of individual noisy features. XGBoost’s larger gain reflects its sensitivity to feature distributions: decision tree splits are chosen based on thresholds of individual features, and noisy features produce splits that capture artifact patterns rather than genuine HRV structure. Filtering tightens the feature distributions and leads to more meaningful splits.

B. Recall Score Comparison

The recall improvements tell a different story from the F1 gains. XGBoost gains the most (+7.93%), followed by MLP (+7.05%), while the SVM gains the least (+2.20%). The SVM, by contrast, gains the most on precision (+8.11%). This inversion suggests that noise affects each classifier’s

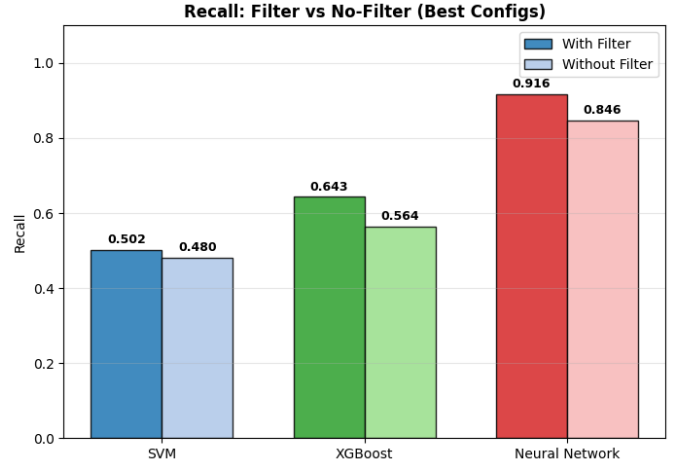


Fig. 10. Recall Comparison with and without filtering for each classifier

error type differently. For XGBoost and MLP, noise appears to corrupt the minority class feature representations: arrhythmia windows become harder to distinguish from baseline, producing false negatives. Filtering restores the signal in these windows and reduces false negatives, boosting recall. For the SVM, the linear decision boundary appears to be more susceptible to noise pushing borderline baseline windows across the boundary into the positive class, generating false positives that filtering eliminates, hence the larger precision gain.

C. Discussion

The consistent improvement across all classifiers confirms that the Butterworth low-pass filter provides a meaningful preprocessing benefit for this task. That said, the improvements are incremental rather than transformative. No classifier crosses a qualitatively different performance threshold with filtering; the relative ranking of the three classifiers on every metric is unchanged. Filtering improves classifiers that are already learning from the data; it does not rescue classifiers that are not.

The differential pattern of gains is practically useful. The SVM’s precision improved more than its recall, meaning filtering makes its conservative tendency more reliable. For XGBoost, the larger recall gain is particularly relevant given the finding in Section 3 that XGBoost offers the most clinically balanced error profile. The filter amplifies this balance, which strengthens the case for using XGBoost with filtering in a deployment scenario. The MLP’s modest gains confirm that when the loss function is already heavily asymmetric, the marginal effect of input noise is smaller.

V. CONCLUSION

This paper evaluated three machine learning classifiers for binary VT/VF detection using HRV features derived from RR intervals, across six hyperparameter configurations each, with and without Butterworth low-pass filtering in the preprocessing pipeline.

The central finding across all three classifiers is that class imbalance was the dominant obstacle to good performance, and that directly addressing it through loss function asymmetry produced larger improvements than any architectural or hyperparameter change. All three best-performing configurations required some form of cost weighting. Among the classifiers, the MLP achieved the highest F1 score (0.6480) and the highest recall (0.9163) with class weighting, but at a precision of 0.5012 that makes it unsuitable for standalone clinical deployment. The SVM achieved the best accuracy (0.6810) and precision (0.7261) but missed approximately half of all VT/VF events. XGBoost offered the most clinically balanced error profile, improving on the SVM's recall while producing fewer false positives than the MLP. Butterworth filtering consistently improved all three classifiers across all metrics, confirming that it provides a meaningful preprocessing benefit. The improvements were incremental rather than transformative, with gains in the range of 2-8 percentage points per metric. The relative ranking of the classifiers was unchanged by filtering. The modest absolute performance of all three classifiers points to a fundamental limitation of HRV-based features alone. Aggregate statistical summaries of RR intervals carry useful discriminative information about VT/VF, but they do not fully capture the arrhythmia signature. Future work should investigate the addition of waveform morphology features from the raw ECG signal, which would provide the classifiers with more discriminative input. A hybrid approach combining HRV and morphological features, evaluated under the same classification framework used here, would provide a direct measure of how much the additional feature set improves detection performance.